

Functional Units of Digital System and Bus Architecture

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1 Introduction to Computer System Architecture

A computer system is an integrated collection of hardware and software components designed to perform computational tasks. The architecture of a computer system defines its structure, organization, and the interconnection between various functional units.

1.1 Basic Computer Organization

A digital computer system consists of several interconnected units that work together to execute instructions and process data. Understanding these functional units and their interconnections is fundamental to computer system architecture.

2 Functional Units of a Digital System

A computer system can be divided into five fundamental functional units:

2.1 Input Unit

The input unit accepts data and instructions from external sources and converts them into a form that the computer can process.

Functions:

- Accepts data and programs from input devices
- Converts input information into binary format
- Transfers data to memory or processor

Examples: Keyboard, mouse, scanner, microphone, touch screen

2.2 Output Unit

The output unit receives processed data from the computer and presents it to the user in human-readable form.

Functions:

- Accepts results from memory or processor

- Converts binary data into human-readable format
- Sends information to output devices

Examples: Monitor, printer, speakers, projector

2.3 Storage Unit (Memory)

The storage unit stores data and instructions. It is divided into two categories:

2.3.1 Primary Memory (Main Memory)

Characteristics:

- Directly accessible by the CPU
- Volatile in nature (RAM)
- Faster access time
- Limited storage capacity
- More expensive per byte

Types:

- **RAM (Random Access Memory):** Volatile, used for temporary storage
- **ROM (Read Only Memory):** Non-volatile, stores permanent instructions
- **Cache Memory:** High-speed memory between CPU and main memory

2.3.2 Secondary Memory

Characteristics:

- Not directly accessible by CPU
- Non-volatile storage
- Slower access time
- Large storage capacity
- Less expensive per byte

Examples: Hard disk, SSD, optical disks, magnetic tapes

2.4 Arithmetic Logic Unit (ALU)

The ALU performs all arithmetic and logical operations.

Arithmetic Operations:

- Addition, Subtraction
- Multiplication, Division
- Increment, Decrement

Logical Operations:

- AND, OR, NOT, XOR
- Complement
- Shift operations
- Comparison operations

2.5 Control Unit (CU)

The control unit coordinates and controls all operations of the computer system.

Functions:

- Fetches instructions from memory
- Decodes instructions
- Generates control signals
- Manages data flow between units
- Controls timing and sequencing

Note: The ALU and Control Unit together form the Central Processing Unit (CPU).

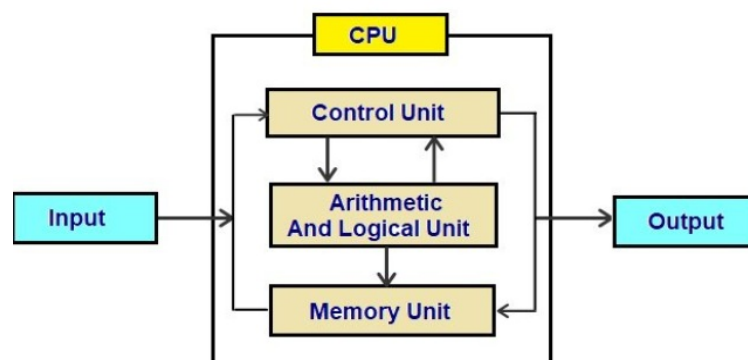


Figure 1: Computer system functional blocks interconnection

3 Interconnection of Functional Units

The functional units of a computer must be interconnected to facilitate data transfer and communication. This interconnection is achieved through a system of buses.

3.1 Need for Interconnection

- Data transfer between processor and memory
- Communication between processor and I/O devices
- Information exchange between different components
- Coordination of operations across the system

4 Computer Buses

A bus is a communication pathway that connects two or more devices. It consists of multiple signal lines, each capable of transmitting one bit of information at a time.

4.1 Bus Characteristics

- **Bus Width:** Number of parallel lines (8-bit, 16-bit, 32-bit, 64-bit)
- **Bus Speed:** Rate of data transfer (measured in MHz or GHz)
- **Bus Protocol:** Set of rules governing communication
- **Bus Master:** Device that initiates bus transactions
- **Bus Slave:** Device that responds to bus master requests

4.2 Advantages of Bus Architecture

- Simplifies system design
- Reduces number of interconnections
- Facilitates modular expansion
- Standardizes communication protocols
- Cost-effective solution

4.3 Disadvantages of Bus Architecture

- Communication bottleneck
- Only one device can use the bus at a time
- Bus length limitations due to signal degradation
- Requires bus arbitration mechanism

5 Types of Buses

Computer buses can be classified based on their function and location in the system.

5.1 Functional Classification

5.1.1 Data Bus

The data bus carries actual data being transferred between components.

Characteristics:

- Bi-directional bus
- Width determines word size (8, 16, 32, 64 bits)
- Carries operands, instructions, and results
- Width directly affects system performance

Example: A 32-bit data bus can transfer 4 bytes simultaneously.

5.1.2 Address Bus

The address bus carries memory addresses or I/O port addresses.

Characteristics:

- Uni-directional bus (CPU to Memory/IO)
- Width determines addressable memory
- n -bit address bus can address 2^n locations
- Does not carry actual data

Example: A 20-bit address bus can address $2^{20} = 1,048,576$ (1 MB) locations.

5.1.3 Control Bus

The control bus carries control signals for coordinating operations.

Characteristics:

- Bi-directional bus
- Variable width
- Carries timing and control signals

Common Control Signals:

- Memory Read (MR)
- Memory Write (MW)
- I/O Read (IOR)

- I/O Write (IOW)
- Interrupt Request (IRQ)
- Bus Request (BR)
- Bus Grant (BG)
- Clock signals
- Reset signals

5.2 Hierarchical Classification

5.2.1 System Bus (Local Bus)

Connects the CPU with main memory and other major components.

Characteristics:

- High-speed bus
- Short physical length
- Directly connected to processor
- Operates at processor speed or fraction thereof

Components:

- Processor bus
- Memory bus
- Cache bus

5.2.2 Expansion Bus (I/O Bus)

Connects peripheral devices and expansion cards to the system.

Characteristics:

- Lower speed compared to system bus
- Longer physical length
- Supports multiple device types
- Standardized interfaces

Examples:

- PCI (Peripheral Component Interconnect)
- PCIe (PCI Express)
- USB (Universal Serial Bus)
- ISA (Industry Standard Architecture)

5.3 Timing Classification

5.3.1 Synchronous Bus

Operations are controlled by a clock signal.

Advantages:

- Simple implementation
- Predictable timing
- Easy to design and test

Disadvantages:

- All devices must operate at same clock speed
- Limited by slowest device
- Clock skew issues

5.3.2 Asynchronous Bus

Operations are controlled by handshaking signals rather than a clock.

Advantages:

- Devices can operate at different speeds
- More flexible
- No clock skew problems

Disadvantages:

- Complex control logic
- Requires handshaking protocol
- Slower due to protocol overhead

6 Bus Architecture

Bus architecture defines how buses are organized and interconnected within a computer system.

6.1 Single Bus Architecture

All components are connected to a single common bus.

Advantages:

- Simple design
- Low cost

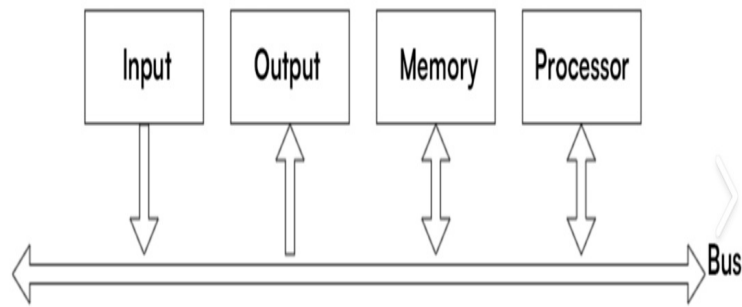


Figure 2: Single bus architecture

- Easy to expand

Disadvantages:

- Performance bottleneck
- Only one transfer at a time
- Bus bandwidth shared by all devices

6.2 Two-Bus Architecture

Separates the system into two buses for improved performance.

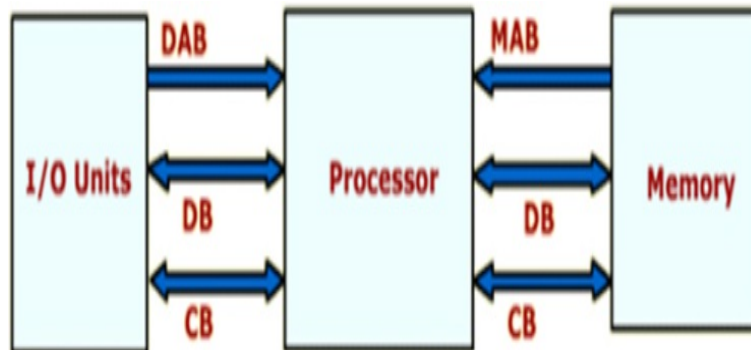


Figure 3: Two bus architecture

Configuration:

- One bus connects CPU to memory
- Second bus connects I/O devices
- Bridge or bus controller connects both buses

Advantages:

- Reduces bus traffic
- CPU-memory transfers independent of I/O
- Better performance

6.3 Three-Bus Architecture

Uses three separate buses for different purposes. **Configuration:**

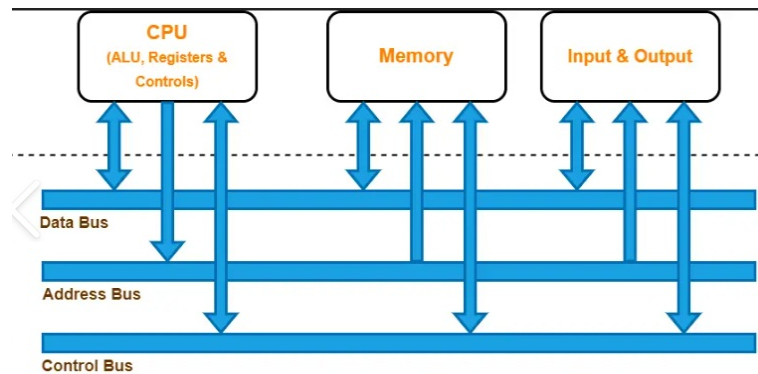


Figure 4: Three bus architecture

- Local bus (CPU-Cache-Memory)
- System bus (Main memory to I/O)
- Expansion bus (Peripheral devices)

Advantages:

- Maximum performance
- Parallel operations possible
- Optimized for different speed requirements

Disadvantages:

- Complex design
- Higher cost
- Requires sophisticated bus controllers

7 Bus Arbitration

When multiple devices share a common bus, a mechanism is needed to determine which device gets access to the bus. This mechanism is called bus arbitration.

7.1 Need for Bus Arbitration

- Multiple devices competing for bus access
- Only one device can use the bus at a time
- Prevents bus conflicts
- Ensures fair access to all devices
- Optimizes bus utilization

7.2 Bus Arbitration Methods

7.2.1 Centralized Arbitration

A single bus controller or arbiter makes all arbitration decisions.

Daisy Chaining Devices are connected in a serial chain for priority resolution.

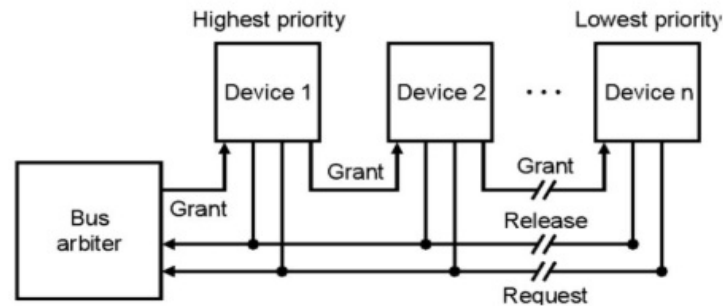


Figure 5: Daisy chain bus arbitration

Working:

- Bus request line is common to all devices
- Bus grant signal is passed through devices in chain
- Device closest to controller has highest priority
- If a device doesn't need the bus, it passes grant to next device

Advantages:

- Simple implementation
- Requires only few control lines
- Easy to add new devices

Disadvantages:

- Fixed priority scheme
- Lower priority devices may face starvation
- Failure of one device affects entire chain
- Priority cannot be changed dynamically

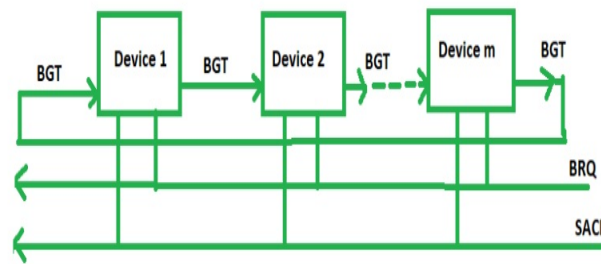


Figure 6: Polling bus arbitration

Polling (Software Approach) Controller polls each device sequentially to check if it needs the bus. **Working:**

- Controller sends device address on bus
- Addressed device responds if it needs bus
- If no response, controller polls next device
- First responding device gets bus access

Advantages:

- Flexible priority assignment
- Can implement various priority schemes
- Easy to modify priorities

Disadvantages:

- Time-consuming process
- Requires address decoding at each device
- Overhead increases with number of devices
- Not suitable for time-critical applications

Independent Request or fixed priority Each device has separate request and grant lines to the controller. **Working:**

- Each device has dedicated request line
- Each device has dedicated grant line
- Controller receives all requests simultaneously
- Priority logic determines which device to grant

Advantages:

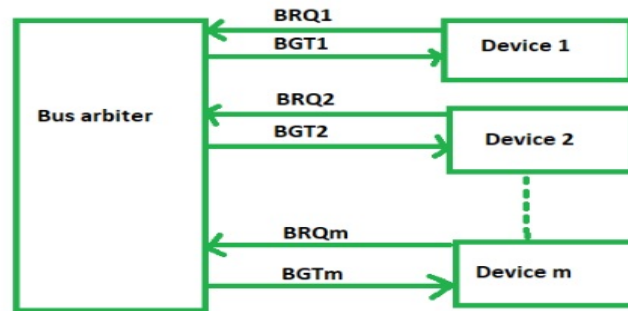


Figure 7: Fixed Priority bus arbitration

- Fast arbitration
- Flexible priority schemes
- Parallel processing of requests
- Most efficient method

Disadvantages:

- Requires many control lines ($2n$ for n devices)
- Complex controller design
- Expensive implementation
- Not scalable for large number of devices

7.2.2 Distributed Arbitration

No central controller; devices cooperate to determine bus access.

Working:

- Each device has arbitration logic
- Devices use arbitration protocol
- Winner is determined by comparing priorities
- No single point of failure

Advantages:

- No central arbitration bottleneck
- More reliable (no single point of failure)
- Faster for small number of devices
- Better scalability

Disadvantages:

- Complex arbitration logic in each device
- Requires more sophisticated devices
- Difficult to implement and debug
- Synchronization challenges

7.3 Priority Arbitration Schemes

7.3.1 Fixed Priority

Each device has a permanently assigned priority level.

Characteristics:

- Simple to implement
- Fast arbitration
- May cause starvation of low-priority devices

7.3.2 Rotating Priority

Priority rotates among devices to ensure fairness.

Characteristics:

- Fair access for all devices
- Prevents starvation
- More complex to implement

7.3.3 Equal Priority (Round Robin)

All devices have equal priority and get access in circular order.

Characteristics:

- Completely fair
- Time-slice based access
- May not be suitable for time-critical devices

8 Data Transfer Techniques

There exists special hardware components between CPU and peripherals to supervise and synchronize all the input and output transfers that are called interface units.

Mode of Transfer: The binary information that is received from an external device is usually stored in the memory unit. CPU merely processes the information but the source and target is always the memory unit. Data transfer between CPU and the I/O devices may be done in different modes:

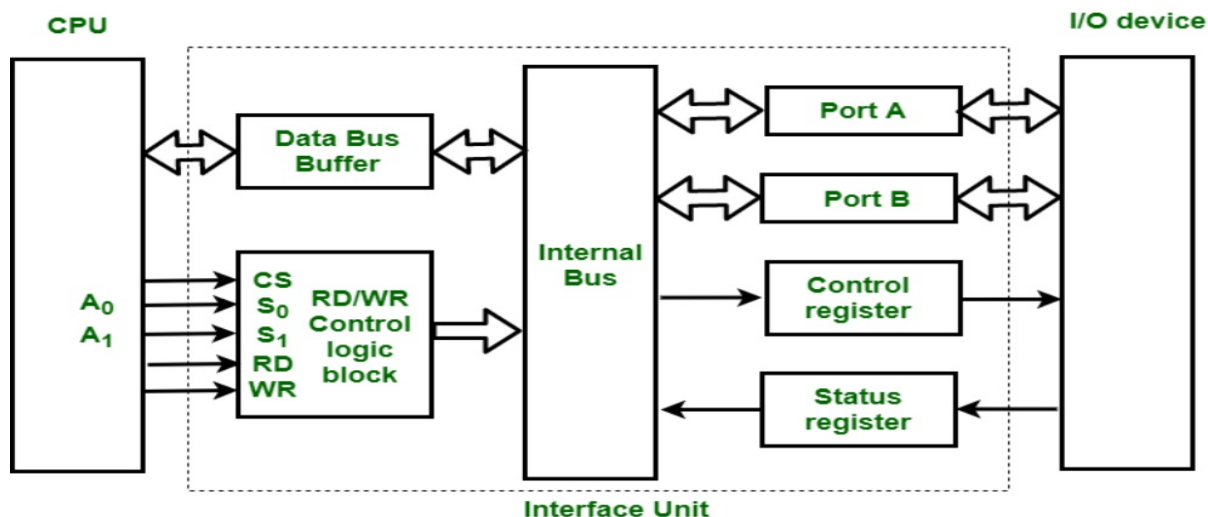


Figure 8: I/O interfacing Unit

8.1 Programmed I/O

CPU executes program to transfer data between memory and I/O device.

Characteristics:

- CPU is fully involved in transfer
- Status checking by CPU
- Simple but inefficient
- CPU time wasted in waiting

8.2 Interrupt-Driven I/O

Device notifies CPU when ready for data transfer through interrupt.

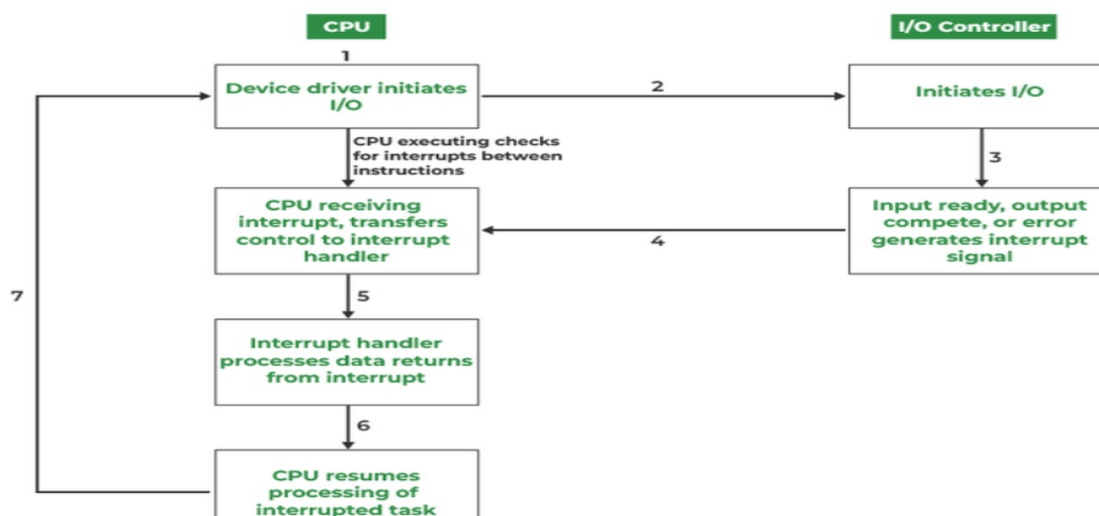


Figure 9: I/O interfacing Unit

Characteristics:

- CPU can perform other tasks while waiting
- Better CPU utilization
- Requires interrupt handling mechanism

8.3 Direct Memory Access (DMA)

Special controller transfers data directly between memory and I/O device without CPU intervention.

Characteristics:

- Fastest data transfer method
- CPU free for other operations
- Requires DMA controller
- Uses bus arbitration to gain bus access

DMA Transfer Steps:

1. CPU initializes DMA controller
2. DMA requests bus from CPU
3. CPU grants bus to DMA (bus arbitration)
4. DMA transfers data
5. DMA sends interrupt when complete

9 Performance Considerations

9.1 Bus Bandwidth

Bus bandwidth is the maximum rate at which data can be transferred over the bus.

Formula:

$$\text{Bandwidth} = \text{Bus Width} \times \text{Bus Clock Frequency}$$

Example: A 64-bit bus operating at 100 MHz has bandwidth:

$$64 \text{ bits} \times 100 \times 10^6 \text{ Hz} = 6.4 \text{ Gbps} = 800 \text{ MBps}$$

9.2 Bus Utilization

Percentage of time the bus is actually transferring data.

Factors Affecting Utilization:

- Bus arbitration delay
- Setup and addressing time
- Handshaking overhead
- Number of competing devices

9.3 Latency

Time delay between bus request and actual data transfer.

Components:

- Arbitration delay
- Bus acquisition time
- Address setup time
- Actual transfer time

10 Modern Bus Standards

10.1 PCI Express (PCIe)

High-speed serial expansion bus standard.

Features:

- Point-to-point serial connections (lanes)
- Scalable bandwidth (x1, x4, x8, x16 lanes)
- Full-duplex communication
- PCIe 4.0: 16 GT/s per lane

10.2 USB (Universal Serial Bus)

Industry standard for peripheral connectivity.

Features:

- Hot-pluggable
- Power supply capability
- USB 3.2: Up to 20 Gbps
- Daisy-chaining support

10.3 SATA (Serial ATA)

Storage device interface standard.

Features:

- Point-to-point connection
- Hot-swapping capability
- SATA III: 6 Gbps
- Simpler cabling than parallel ATA

10.4 Important Formulas

- Addressable Memory: 2^n locations for n -bit address bus
- Bus Bandwidth: Bus Width $\times$ Clock Frequency
- Data Transfer Time: Bus Width $\div$ Bandwidth

11 Practice Questions

11.1 Short Answer Questions

1. Define the five functional units of a computer system.
2. What is a bus? Explain its need in computer architecture.
3. Differentiate between data bus, address bus, and control bus.
4. What is bus arbitration? Why is it necessary?
5. Compare synchronous and asynchronous buses.

11.2 Long Answer Questions

1. Explain the interconnection of functional units in a computer system with a neat diagram.
2. Describe different types of buses based on function and hierarchy.
3. Explain centralized bus arbitration techniques in detail.
4. Compare single-bus, two-bus, and three-bus architectures.
5. Discuss the role of DMA in data transfer and its relation to bus arbitration.

11.3 Numerical Problems

1. A computer has a 32-bit address bus. Calculate the maximum addressable memory.
2. A system uses a 64-bit data bus operating at 200 MHz. Calculate the maximum bandwidth in MBps.
3. If a bus transfer takes 4 clock cycles and the clock frequency is 100 MHz, what is the transfer time?

References

M. Morris Mano (2006), *Computer System Architecture*, 3rd Edition, Pearson/PHI, India.
